

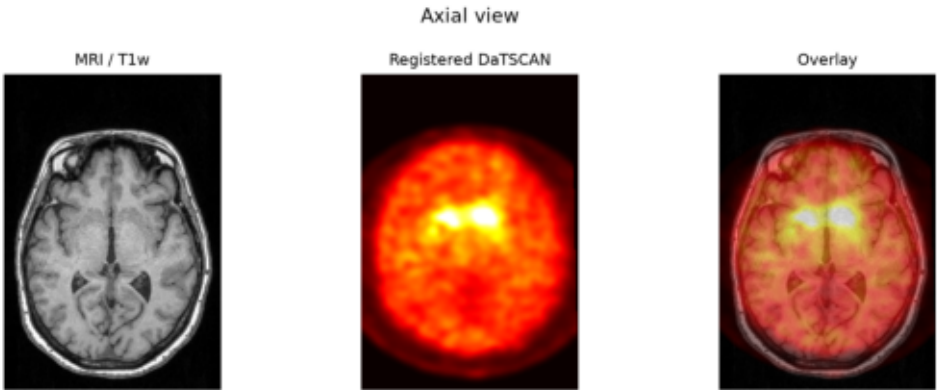
DARQ report

Subject: tutorial_subject
Report date: 2026-06-20 13:05

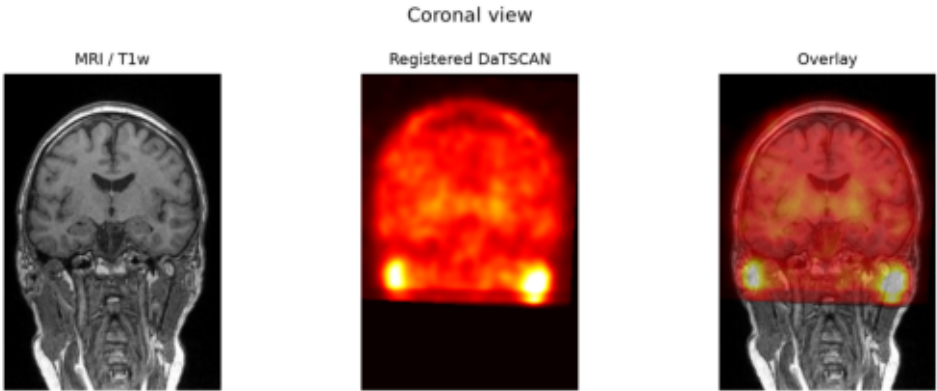
DaTSCAN quantification with MRI/T1w and SynthSeg

Main formulas: $SBR = \text{regional uptake} / \text{occipital uptake} - 1$; $AI = 100 \times (\text{Right} - \text{Left}) / \text{mean}(\text{Right}, \text{Left})$.

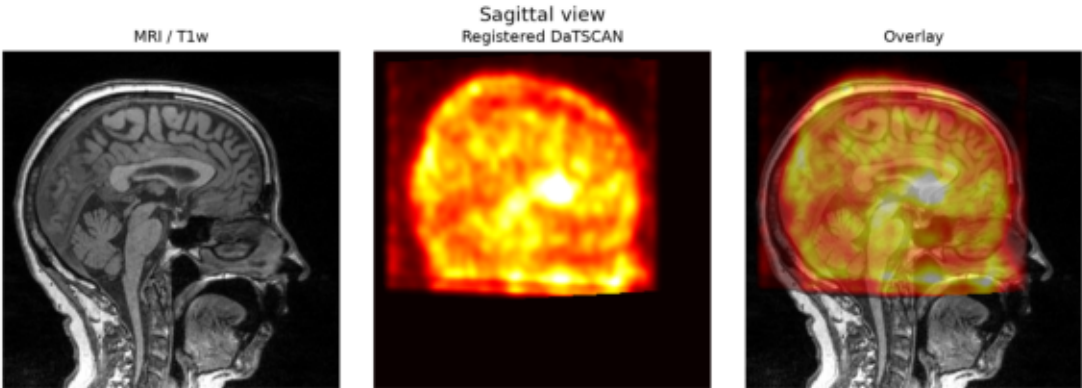
Axial



Coronal



Sagittal



Regional SBR summary

Region	Right SBR	Left SBR	Bilateral SBR	Asymmetry index (%)
Striatum	1.2125	1.3222	1.2669	-8.6589
Caudate	1.451	1.3499	1.4026	7.2229
Putamen	1.0379	1.3044	1.174	-22.7531

Hemisphere SBR table

Hemisphere	SBR striatum	SBR caudate	SBR putamen	Occipital reference	Registration loss
R	1.2125	1.451	1.0379	5734.8774	0.3191
L	1.3222	1.3499	1.3044	5633.6621	0.3191
Both	1.2669	1.4026	1.174	5685.0737	0.3191

Symmetry summary

Metric	Hemisphere	Caudate	Putamen
Incc	R	0.0056	0.0
Incc	L	0.3263	0.1791
Incc	Both	0.1615	0.0929
I2	R	2.6362	2.1919
I2	L	1.4446	1.1257
I2	Both	2.0573	1.6389

Execution log

 Pipeline ejecutado correctamente.

Carpeta de salida:
C:\Users\Usuario\Desktop\Practicas_DaTSCAN\darq\gradio_runs\run_20260620_125806_tutorial_subject\output

Filas SBR encontradas: 21
Filas de simetría encontradas: 30

Última salida de terminal:
ithm #

Total sessions to process: N=1

Subject: 0 (0/1)
* Preprocessing.
* DaT symmetry and mask.
* MRI DaT simulation and mask.
* MRI-DaT registration.
Optimizer: LBFGS lr=0.1 max_iter=10
Losses: reg(w=1); reg_label(w=2); reg_uptake(w=0.0); reg_lr(w=0.5); regularizer(w=1)

Training started #

* Epoch: 0
Epoch summary: loss_reg: 0.355,w_loss_reg: 0.355,loss_reg_label: 0.001,w_loss_reg_label: 0.002,loss_reg_lr: 0.0,w_loss_reg_lr: 0.0,loss_regularizer: 0.0,w_loss_regularizer: 0.0,loss: 0.357,time_duration (s): 21.74

* Epoch: 1
Epoch summary: loss_reg: 0.346,w_loss_reg: 0.346,loss_reg_label: 0.001,w_loss_reg_label: 0.003,loss_reg_lr: 0.001,w_loss_reg_lr: 0.0,loss_regularizer: 0.002,w_loss_regularizer: 0.002,loss: 0.35,time_duration (s): 18.019

* Epoch: 2
Epoch summary: loss_reg: 0.339,w_loss_reg: 0.339,loss_reg_label: 0.002,w_loss_reg_label: 0.004,loss_reg_lr: 0.001,w_loss_reg_lr: 0.0,loss_regularizer: 0.001,w_loss_regularizer: 0.001,loss: 0.344,time_duration (s): 16.203

* Epoch: 3
Epoch summary: loss_reg: 0.312,w_loss_reg: 0.312,loss_reg_label: 0.005,w_loss_reg_label: 0.01,loss_reg_lr: 0.001,w_loss_reg_lr: 0.0,loss_regularizer: 0.001,w_loss_regularizer: 0.001,loss: 0.323,time_duration (s): 16.118

* Epoch: 4
Epoch summary: loss_reg: 0.309,w_loss_reg: 0.309,loss_reg_label: 0.005,w_loss_reg_label: 0.01,loss_reg_lr: 0.0,w_loss_reg_lr: 0.0,loss_regularizer: 0.001,w_loss_regularizer: 0.001,loss: 0.32,time_duration (s): 16.007

* Epoch: 5
Epoch summary: loss_reg: 0.309,w_loss_reg: 0.309,loss_reg_label: 0.005,w_loss_reg_label: 0.009,loss_reg_lr: 0.0,w_loss_reg_lr: 0.0,loss_regularizer: 0.0,w_loss_regularizer: 0.0,loss: 0.319,time_duration (s): 16.178

* Epoch: 6
Epoch summary: loss_reg: 0.309,w_loss_reg: 0.309,loss_reg_label: 0.005,w_loss_reg_label: 0.01,loss_reg_lr: 0.0,w_loss_reg_lr: 0.0,loss_regularizer: 0.0,w_loss_regularizer: 0.0,loss: 0.319,time_duration (s): 17.457

* Epoch: 7
Epoch summary: loss_reg: 0.309,w_loss_reg: 0.309,loss_reg_label: 0.005,w_loss_reg_label: 0.01,loss_reg_lr: 0.0,w_loss_reg_lr: 0.0,loss_regularizer: 0.0,w_loss_regularizer: 0.0,loss: 0.319,time_duration (s): 17.428

* Epoch: 8
Epoch summary: loss_reg: 0.309,w_loss_reg: 0.309,loss_reg_label: 0.005,w_loss_reg_label: 0.01,loss_reg_lr: 0.0,w_loss_reg_lr: 0.0,loss_regularizer: 0.0,w_loss_regularizer: 0.0,loss: 0.319,time_duration (s): 16.099

* Epoch: 9
Epoch summary: loss_reg: 0.309,w_loss_reg: 0.309,loss_reg_label: 0.005,w_loss_reg_label: 0.01,loss_reg_lr: 0.0,w_loss_reg_lr: 0.0,loss_regularizer: 0.0,w_loss_regularizer: 0.0,loss: 0.319,time_duration (s): 16.058

Training finished #

Final loss: 0.31907401164448074
Done in 425.68s.

Failed subjects: 0 / 1
(none)

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All DONE #
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Errores / warnings:
"clear" no se reconoce como un comando interno o externo,
programa o archivo por lotes ejecutable.
C:\Users\Usuario\Desktop\Practicas_DaTSCAN\darq\darq\processing.py:76: FutureWarning: `binary\_opening` is deprecated since version 0.26 and will be removed in version 0.28. Use `skimage.morphology.opening` instead.
mask = binary_opening(mask, ball(3))
C:\Users\Usuario\Desktop\Practicas_DaTSCAN\darq\darq\processing.py:340: FutureWarning: `binary\_dilation` is deprecated since version 0.26 and will be removed in version 0.28. Use `skimage.morphology.dilation` instead. Note the lack of mirroring for non-symmetric footprints (see docstring notes).
mask_dilated = binary_dilation(mask_symm, np.ones((10, 10, 10)))
C:\Users\Usuario\Desktop\Practicas_DaTSCAN\darq\darq\src\preprocessing.py:63: FutureWarning: `binary\_opening` is deprecated since version 0.26 and will be removed in version 0.28. Use `skimage.morphology.opening` instead.
image = binary_opening(image, self.struct_fn(data_dict[k]))